

Research Report: Evaluation of Agentic Signals for Volatility Prediction

Stock: Nvidia (NVDA) Period: 2019

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Abstract

This report evaluates whether agentic signals produced by a multi-agent large language model (LLM) debate can predict next-day realized volatility for Nvidia (NVDA) during the 2019 post-crypto-crash recovery period. We test the primary hypothesis that *agent disagreement* (the cross-agent dispersion of daily scores) leads volatility. The hypothesis is rejected: disagreement exhibits near-zero correlation with realized volatility and is statistically insignificant. A secondary signal emerges: *aggregate agent sentiment* (the mean daily agent score) is significantly negatively correlated with realized volatility and behaves like a discrete “risk-off” detector. Incorporating the mean score into a continuous GARCH-X model yields only marginal out-of-sample accuracy improvements and a worse AIC, indicating the signal is more useful as a threshold-based decision rule than as a continuous variance regressor.

Keywords: multi-agent LLMs; volatility forecasting; realized volatility; GARCH-X; risk-off signals.

1 Executive Summary

This study evaluates the efficacy of multi-agent LLM debates in predicting realized volatility for Nvidia (NVDA) over 252 trading days in 2019.

Primary hypothesis (rejected). Agent disagreement (dissonance among agents) acts as a leading indicator of volatility. Statistical analysis rejects this hypothesis ($p > 0.05$).

Secondary finding (supported). Aggregate agent sentiment (mean score) exhibits a statistically significant negative correlation with realized volatility ($p < 0.001$), consistent with a “risk-off” mechanism: when agents converge to bearish sentiment, volatility tends to expand in the subsequent session.

Model validation. Integrating the mean score into a continuous GARCH-X model reduces RMSE by only 0.11 % relative to a baseline GARCH(1,1), while AIC worsens, suggesting the added parameterization is not justified for continuous variance prediction.

2 Methodology

2.1 Experimental Design

We simulate a daily debate between three specialized AI agents over 252 trading days in 2019, producing one scalar score per agent per day. Agents follow a *Blind & Battle* protocol intended to reduce consensus-seeking bias: agents score independently, then exchange arguments, and may revise their scores.

2.2 Agents

- **Sentiment Agent:** summarizes the 24-hour news cycle (e.g., crypto mining, data center demand).
- **Macro Agent:** analyzes macro proxies (VIX, 10Y yields) and cross-asset relationships (e.g., Bitcoin).
- **Technical Agent:** analyzes daily price momentum and candlestick-style patterns.

2.3 Variables

Let $s_t^{(i)}$ be the score of agent $i \in \{1, 2, 3\}$ on day t .

Aggregate sentiment (mean score).

$$\mu_t = \frac{1}{3} \sum_{i=1}^3 s_t^{(i)}.$$

Agent disagreement (confidence dispersion).

$$D_{\text{conf},t} = \sqrt{\frac{1}{3-1} \sum_{i=1}^3 \left(s_t^{(i)} - \mu_t\right)^2}.$$

Target variable: next-day realized volatility. We use next-day realized volatility computed from log returns; denote it σ_{t+1}^{RV} .

2.4 Statistical Tests

We evaluate linear association using Pearson correlation between each signal ($D_{\text{conf},t}$ and μ_t) and next-day realized volatility σ_{t+1}^{RV} .

3 Results I: Rejection of Agent Disagreement

We hypothesized that elevated disagreement $D_{\text{conf},t}$ would precede volatility spikes. Empirically, periods of high disagreement more frequently coincided with sideways price action (consolidation) rather than volatility expansion.

Statistical verdict. The correlation between disagreement and next-day realized volatility is near zero and statistically insignificant.

4 Results II: Discovery of a “Consensus Fear” Signal

After the primary hypothesis failed, we analyzed the mean score μ_t as a candidate predictor.

Observation. When agents achieved a bearish consensus (e.g., $\mu_t < -0.5$), volatility tended to expand in the subsequent session.

Statistical verdict. Mean score exhibits a strong inverse relationship with next-day realized volatility.

5 Statistical Scorecard

Table 1: Correlation tests between agentic signals and next-day realized volatility for NVDA in 2019.

Signal	Pearson r	p -value
Agent disagreement (D_{conf})	-0.0727	0.2557
Mean score (μ)	-0.2481	0.0001

6 GARCH-X Model Validation

We validate whether the mean score improves volatility forecasting when used as an exogenous regressor.

6.1 Models

Baseline: GARCH(1,1). Let ϵ_t be returns residuals with conditional variance h_t .

$$h_t = \omega + \alpha\epsilon_{t-1}^2 + \beta h_{t-1}.$$

Augmented: GARCH-X. Incorporate mean score μ_{t-1} as an exogenous term:

$$h_t = \omega + \alpha\epsilon_{t-1}^2 + \beta h_{t-1} + \gamma \mu_{t-1}.$$

6.2 Results

Table 2: Model comparison for volatility forecasting (RMSE, AIC).

Model	RMSE	AIC
Standard GARCH(1,1)	0.019908	843.97
Agent GARCH-X (with μ)	0.019886	844.44

Interpretation. While RMSE improves by 0.11 %, AIC worsens, suggesting the additional complexity is not justified for a continuous variance regressor. Operationally, μ_t appears better suited to a binary “risk-off” threshold.

7 Conclusion & Strategic Pivot

Key findings.

1. Agent disagreement produces noise rather than predictive signal.
2. Agent consensus on negative sentiment (low mean score) is a strong predictor of subsequent volatility expansion.

Strategic pivot. Future work should shift from continuous GARCH-X regression to a binary threshold strategy (e.g., if $\mu_t < -0.5$, go risk-off), which better matches the discrete nature of the panic/consensus mechanism.

8 Figures

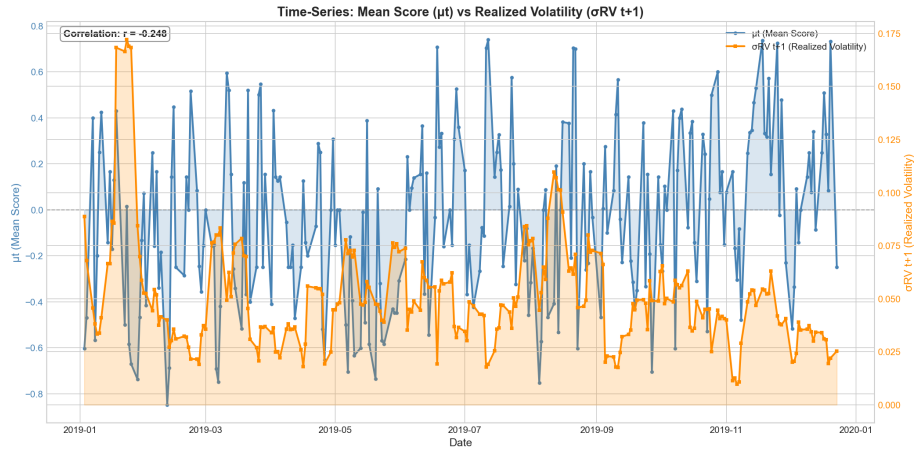


Figure 1: Time series of aggregate agent sentiment μ_t and next-day realized volatility σ_{t+1}^{RV} for NVDA (2019).

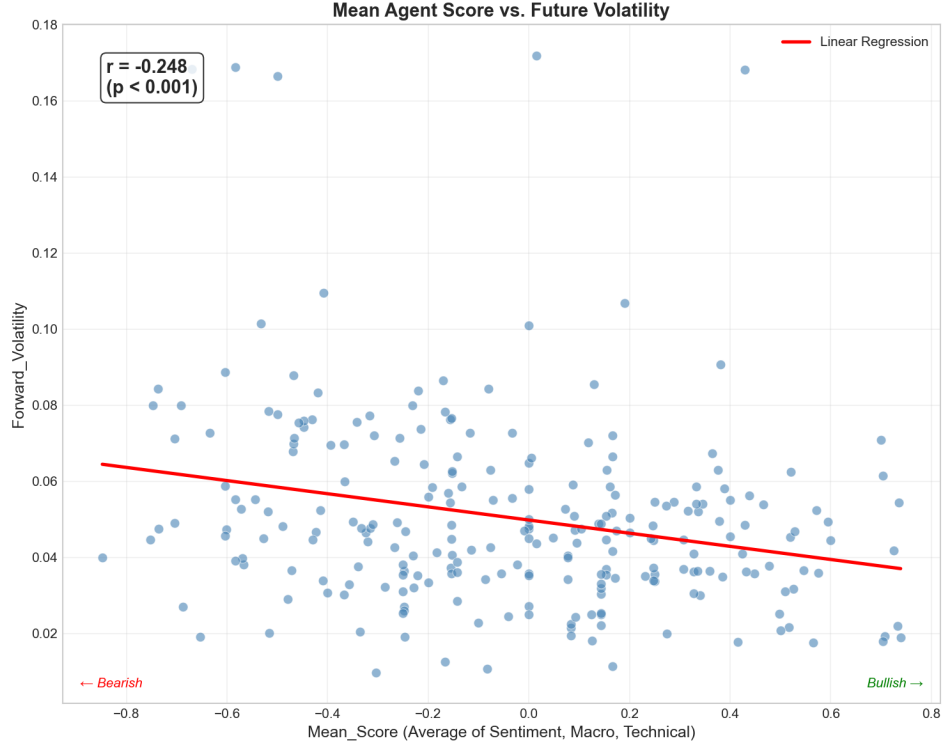


Figure 2: Scatter plot of mean agent score μ_t versus next-day realized volatility σ_{t+1}^{RV} .

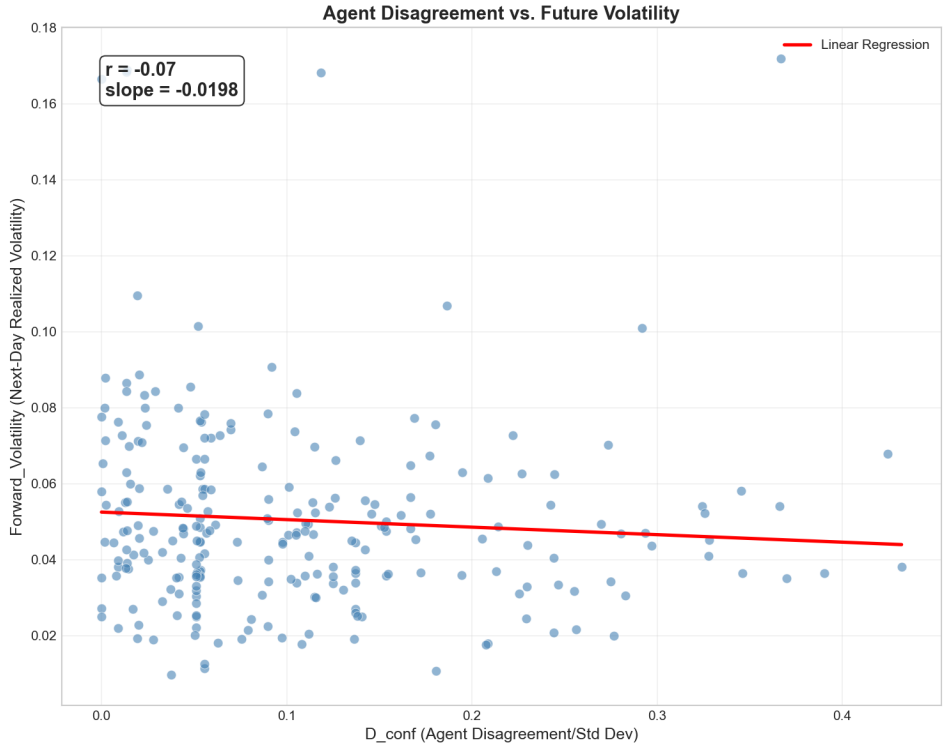


Figure 3: Scatter plot of agent disagreement $D_{\text{conf},t}$ versus next-day realized volatility σ_{t+1}^{RV} .

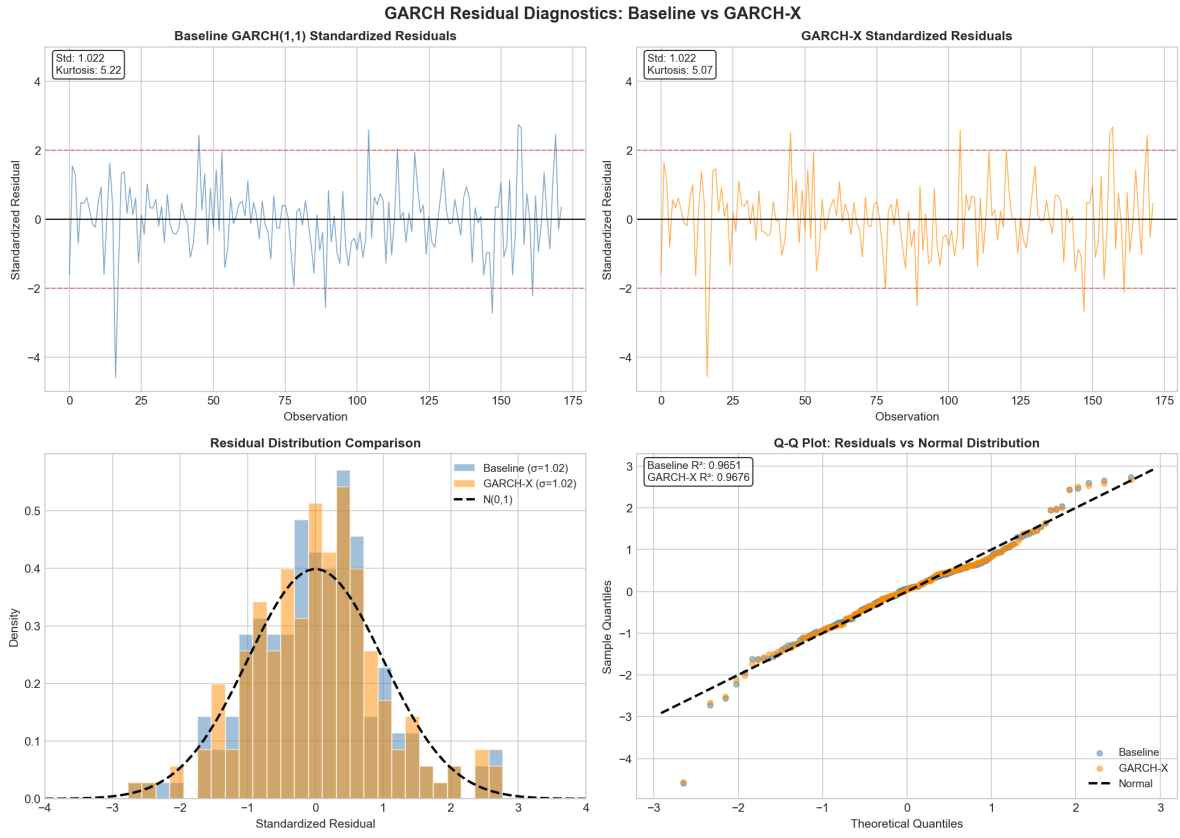


Figure 4: Residual diagnostics (illustrative) for baseline GARCH(1,1) versus Agent GARCH-X.